



AI ENGINEER ROADMAP

EVERY AI ENGINEER SHOULD KNOW THESE **CONCEPTS**



HOW IT WORKS



WHY IT WORKS



WHERE IT FAILS



HOW TO BUILD IT



Not just how to use AI.

But how it works, why it works, where it fails,
and how to build it in production.

10 CONCEPTS →

Swipe through the roadmap →

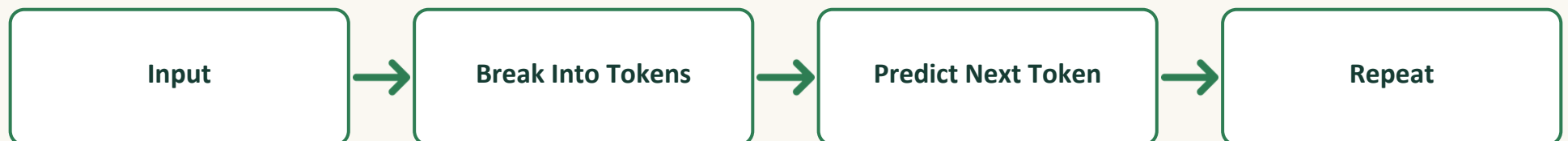
Large Language Models

(Predicting the Next Word)

WHAT IT IS

An LLM is trained on massive amounts of text to predict the next word, like a widely-read student answering from everything it's seen.

HOW IT WORKS



EXAMPLE

Ask “Write a job description” and it generates “We are looking for a Full Stack Developer...”

WHERE IT FAILS

- ✗ Hallucinations
- ✗ Outdated knowledge
- ✗ No access to your private data



KEY INSIGHT

It predicts patterns — it doesn't know facts.

That's why grounding it with real data (retrieval) matters more than a bigger model.

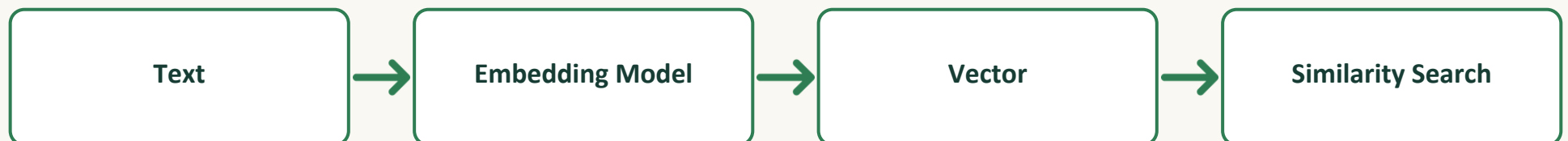
Embeddings

(Turning Meaning Into Numbers)

WHAT IT IS

Embeddings convert text into vectors of numbers that capture meaning — computers match numbers, not words.

HOW IT WORKS



EXAMPLE

Searching “cheap phone” also surfaces “affordable smartphone,” even though the words differ.

WHERE IT FAILS

- ✗ Very similar meanings can sometimes confuse retrieval



KEY INSIGHT

Meaning is compared, not words.

This one idea is what makes semantic search possible at all.

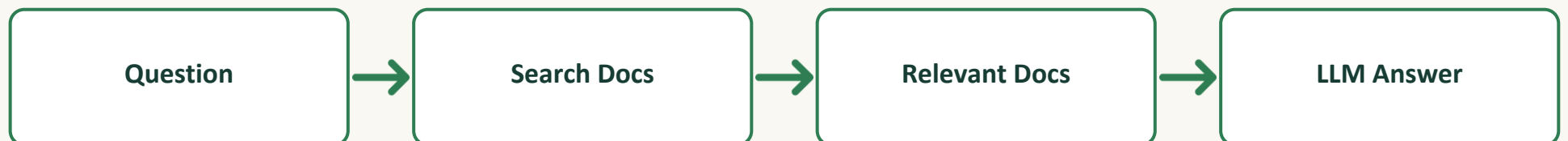
Retrieval

(Search Before You Answer)

WHAT IT IS

Retrieval finds the most relevant information before the LLM answers, instead of asking it to remember everything.

HOW IT WORKS



EXAMPLE

A question is matched against company documents, and only the relevant pages are handed to the model.

WHERE IT FAILS

✗ Wrong document retrieved → wrong answer



KEY INSIGHT

Search first, then answer.

Retrieval quality is often the real bottleneck — not the LLM itself.

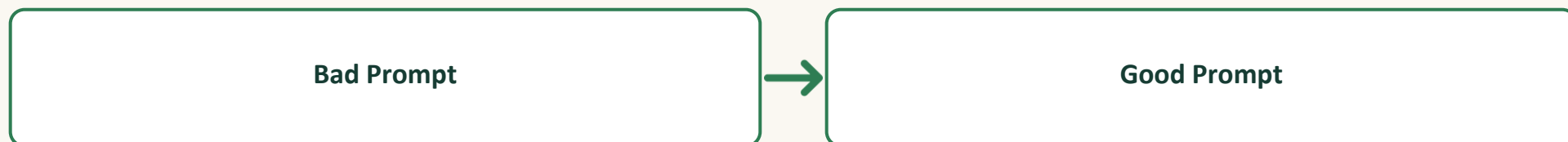
Prompt Engineering

(Better Instructions, Better Output)

WHAT IT IS

The prompt is simply the instructions you give the AI — LLMs rely heavily on that context to respond well.

HOW IT WORKS



EXAMPLE

“Write code” vs. “Write a React login component using TypeScript and Tailwind.”

WHERE IT FAILS

✖ An ambiguous prompt produces poor output



KEY INSIGHT

The instructions are the interface.

A better prompt is usually cheaper than a bigger model.

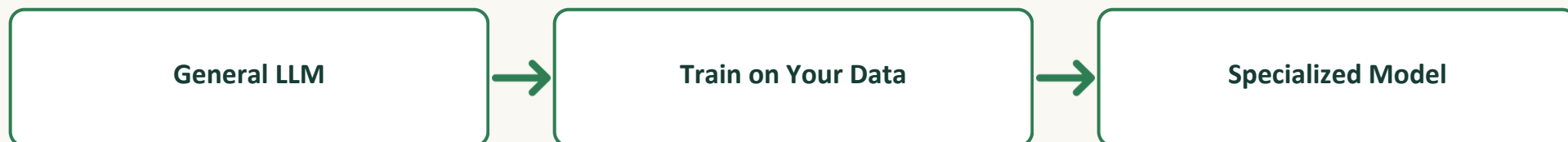
Fine-Tuning

(Teaching a Model New Behavior)

WHAT IT IS

Fine-tuning trains an existing model further on your own data, for when prompting alone isn't enough.

HOW IT WORKS



EXAMPLE

A general LLM trained on legal contracts becomes a capable legal assistant.

WHERE IT FAILS

- ✗ Expensive
- ✗ Needs lots of quality data
- ✗ Can overfit



KEY INSIGHT

Reach for it last, not first.

Most problems are solved with better prompts or retrieval before this is ever needed.

RAG

(Retrieval-Augmented Generation)

WHAT IT IS

RAG lets an LLM search external knowledge before answering, keeping responses accurate and current.

HOW IT WORKS



EXAMPLE

Chatting with your own company documents instead of the model's general knowledge.

WHERE IT FAILS

✖ Poor retrieval → wrong answer



KEY INSIGHT

RAG buys freshness and accuracy.

But it's only ever as good as what actually gets retrieved.

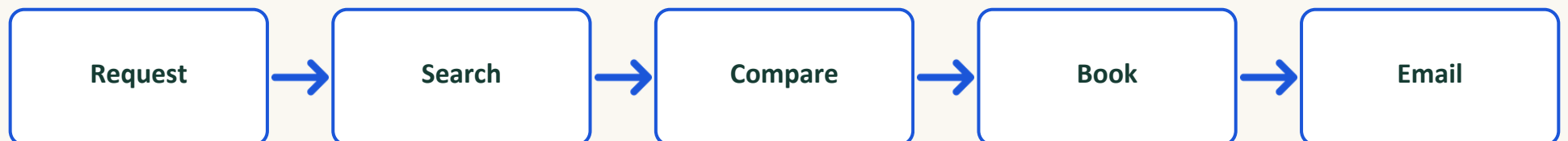
Agent Workflows

(A Checklist, Not a Single Answer)

WHAT IT IS

Instead of answering directly, the AI performs multiple steps — like giving it a checklist instead of one question.

HOW IT WORKS



EXAMPLE

“Book my trip” triggers searching flights, comparing prices, booking a hotel, and sending an email.

WHERE IT FAILS

- ✗ Too many steps raise complexity and cost



KEY INSIGHT

Give it a checklist, not a question.

More autonomy means more places for things to quietly go wrong.

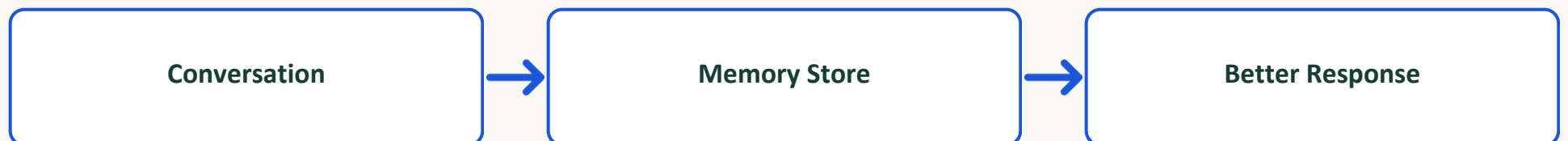
Memory Systems

(Remembering Across Conversations)

WHAT IT IS

Memory lets AI carry useful information across a conversation or task, instead of starting from scratch each time.

HOW IT WORKS



EXAMPLE

Tell it “I’m a frontend developer,” and later it tailors a learning path around that.

WHERE IT FAILS

- ✗ Forgets details
- ✗ Keeps outdated info
- ✗ Context grows too large



KEY INSIGHT

No memory means starting from zero, every time.

The hard part isn't storing memory — it's deciding what's worth keeping.

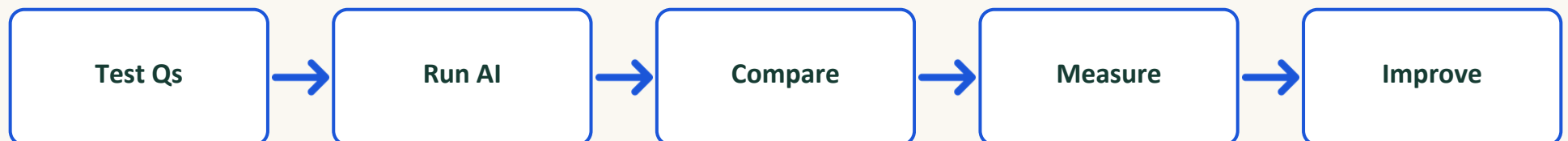
Evaluation Pipelines

(Measuring Quality, Not Guessing)

WHAT IT IS

A repeatable process for testing whether your AI system is actually producing high-quality answers.

HOW IT WORKS



EXAMPLE

Run 100 known questions and check whether the system retrieves and answers correctly.

WHERE IT FAILS

✗ Relying on “it looks good” instead of measurable results



KEY INSIGHT

If you can't measure it, you can't improve it.

Gut feeling isn't a metric — and it doesn't scale with your team.

Production Deployments

(From Laptop to Real Users)

WHAT IT IS

Making an AI application reliable, scalable, and secure for real users — not just a working demo.

HOW IT WORKS



EXAMPLE

A customer support chatbot used by thousands of people every day.

WHERE IT FAILS

- ✗ Works perfectly on your laptop, crashes under real traffic



KEY INSIGHT

Working on your laptop isn't the finish line.

Real traffic finds every weak spot instantly — performance, cost, and error handling included.